

Quiz 3 (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Matrix Algebra

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -4 \\ 6 & 4 \end{bmatrix} \quad \vec{b}_1, \vec{b}_2$$

Determine a matrix X such that $B^{-1}AX = I$, where I is the identity matrix.

$$B^{-1}AX = I \Rightarrow AX = B \Rightarrow X = [\vec{x}_1, \vec{x}_2] \text{ and } A\vec{x}_1 = \vec{b}_1, A\vec{x}_2 = \vec{b}_2$$

$$A\vec{x}_1 = \vec{b}_1:$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad \begin{array}{l} x_1 = -2 + x_3 \\ x_2 = 3 - 2x_3 \\ x_3 \text{ is free} \end{array}$$

$$A\vec{x}_2 = \vec{b}_2:$$

$$\begin{bmatrix} 1 & 1 & 1 & -4 \\ 0 & 2 & 4 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & -4 \\ 0 & 1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & -6 \\ 0 & 1 & 2 & 2 \end{bmatrix} \quad \begin{array}{l} x_1 = -6 + x_3 \\ x_2 = 2 - 2x_3 \\ x_3 \text{ is free} \end{array}$$

$$x_1 = \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} -6 \\ 2 \\ 0 \end{bmatrix}$$

* Note: any specific solutions for the given general form solutions work

Solution.

$$\begin{bmatrix} -2 & -6 \\ 3 & 2 \\ 0 & 0 \end{bmatrix}$$

2 Linear Dependence

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ 7 \\ -5 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ k \\ 1 \end{bmatrix}$$

Determine the value of k such that the above set of vectors is linearly dependent.

$$\begin{bmatrix} 1 & -4 & 2 \\ 2 & 7 & k \\ -1 & -5 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 2 \\ 0 & 15 & k-4 \\ 0 & -9 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & -1 \\ 0 & 15 & k-4 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & k+1 \end{bmatrix}$$

$k = -1 \Rightarrow$ too few pivots

Solution.

$$k = -1$$

3 Linear Transformations

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 - x_2 + 4x_3 \\ 3x_1 + 10x_2 - x_3 \\ -x_1 - 5x_2 + 2x_3 \end{bmatrix}$$

Determine if the above linear transformation is (1) one-to-one and not onto (2) onto and not one-to-one (3) both one-to-one and onto or (4) neither one-to-one nor onto. Justify your answer

$$\begin{bmatrix} 1 & -1 & 4 \\ 3 & 10 & -1 \\ -1 & -5 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 \\ 0 & 13 & -13 \\ 0 & -6 & 6 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & -1 & 4 \\ 0 & \boxed{1} & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Solution.

neither. Too few pivots per row and per column